

PBD9 Chapter 20 and PBD8 Chapter 20: The Kidney

BP9 Chapter 13 and BP8 Chapter 14: The Kidney and Its Collecting System

1 A 71-year-old man has had decreased urine output <500 mL per day for the past 3 days. Physical examination shows vital signs with temperature 37° C, pulse 88/min, respiratory rate 18/min, and blood pressure 85/60 mm Hg. He has peripheral edema and diffuse rales on auscultation of the chest. Urinalysis shows specific gravity 1.019 and no protein, blood, glucose, ketones, WBCs, RBCs, or casts. His serum creatinine is 3.3 mg/dL, and urea nitrogen is 62 mg/dL. The fractional excretion of sodium (FE_{Na}) is <1%. Which of the following underlying conditions is he most likely to have?

- A Dilated cardiomyopathy
- B Membranous nephropathy
- C Prostatic hyperplasia
- D Systemic lupus erythematosus
- E Urothelial carcinoma

2 A 36-year-old woman has had increased malaise for 3 weeks and urine output <500 mL/day for the past 4 days. On examination, she has blood pressure 170/112 mm Hg and peripheral edema. Urinalysis shows protein 1+ and blood 3+, but no glucose or ketones. Urine microscopic analysis shows RBCs and RBC casts. Her serum urea nitrogen is 39 mg/dL, and creatinine is 4.3 mg/dL. Her serum complement C1q, C3, and C4 are decreased. A renal biopsy is performed, and immunofluorescence microscopy shows a granular pattern of staining with antibody to C3. Which of the following types of hypersensitivity reactions is most likely causing her renal disease?

- A I (IgE-mediated systemic anaphylaxis)
- B II (Antibody-dependent cell-mediated cytotoxicity)
- C III (Immune complex formation)
- D IV (Delayed-type hypersensitivity)

3 A 29-year-old man with chronic hepatitis C virus infection has noted dark urine for the past 2 weeks. On examination he is hypertensive but afebrile. Laboratory studies show serum creatinine of 3.8 mg/dL and urea nitrogen of 35 mg/dL.

Cryoglobulins are detected. Urinalysis shows RBCs and RBC casts. A renal biopsy is performed and microscopically shows hypercellular glomeruli with lobulation and a double-contour appearance to split basement membranes adjacent to sub-endothelial immune complexes. Which of the following cell types has most likely proliferated in his glomeruli?

- A Juxtaglomerular cells
- B Mesangial cells
- C Parietal epithelial cells
- D Podocytes
- E Endothelial cells

4 A study of renal disease identifies patients with greater than 3.5 g of protein in a 24-hour urine collection, but no RBCs or WBCs. Dysfunction involving which of the following cells is most likely to be responsible for proteinuria?

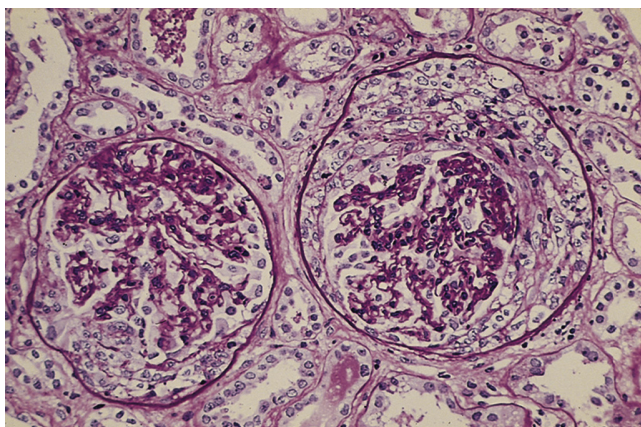
- A Endothelium
- B Macula densa
- C Mesangium
- D Parietal epithelium
- E Podocytes

5 A 7-year-old boy is recovering from impetigo. Physical examination shows five honey-colored crusts on his face. The crusts are removed, and a culture of the lesions grows *Streptococcus pyogenes*. He is treated with antibiotics. One week later, he develops malaise with nausea and a slight fever and passes dark brown urine. Laboratory studies show a serum anti-streptolysin O titer of 1:1024. Which of the following is the most likely outcome of his renal disease?

- A Chronic renal failure
- B Complete recovery
- C Crescentic glomerulonephritis
- D Rheumatic heart disease
- E Streptococcal urinary tract infection

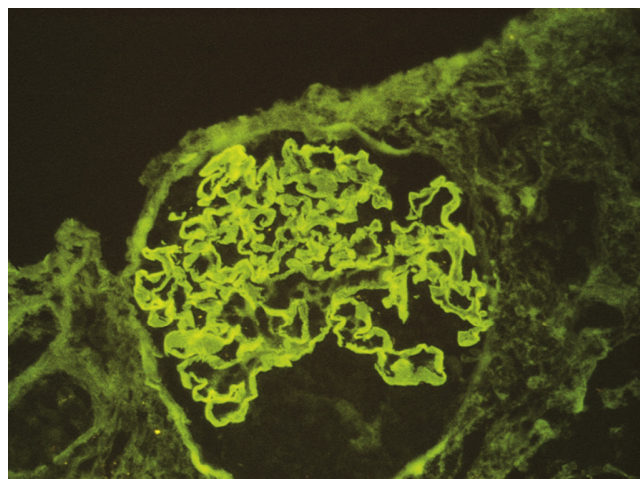
6 A 17-year-old girl living in the Congo has had a chronic febrile illness for 2 years. In the past 2 days she notes her urine is smoky brown. On physical examination her blood pressure is 145/95 mm Hg. Laboratory studies show her serum creatinine is 3.7 mg/dL, and urea nitrogen is 35 mg/dL. Urinalysis shows 4+ blood with 1+ protein, but no glucose, ketones, or leukocytes. The serum haptoglobin is decreased and Coombs test is negative. Her Hgb is 8.5 g/dL. A peripheral blood smear shows rare ring stage trophozoites. Immunofluorescence microscopy performed on renal biopsy shows granular deposition of IgG and C3 in glomerular capillary basement membranes. Electron microscopy shows electron-dense sub-epithelial "humps." Which of the following renal diseases is she most likely to have?

- A Focal segmental glomerulosclerosis
- B Hereditary nephritis
- C IgA nephropathy
- D Lupus nephritis
- E Membranous nephropathy
- F Acute proliferative glomerulonephritis



7 A 45-year-old man has experienced increasing malaise, nausea, and reduced urine output for the past 3 days. On physical examination, he is afebrile and normotensive. Laboratory findings show a serum creatinine level of 7.5 mg/dL. Urinalysis shows hematuria, but no pyuria or glucosuria. A renal biopsy is done; the light microscopic picture is shown in the figure. Which of the following additional studies is most useful for classification and treatment of this disease?

- A Antinuclear antibody
- B Anti-glomerular basement membrane antibody
- C HIV-1 RNA copy level
- D Quantitative serum immunoglobulins
- E Rheumatoid factor
- F Urine immunoelectrophoresis



8 A 21-year-old previously healthy man has noticed blood in his urine for the past 2 days. He reports no dysuria, frequency, or hesitancy of urination. On physical examination, there are no abnormal findings. Laboratory findings show a serum urea nitrogen level of 39 mg/dL and creatinine level of 4.1 mg/dL. A renal biopsy specimen is obtained; the immunofluorescence pattern of staining with antibody against human IgG is shown in the figure. Which of the following serologic findings is most likely to be present in this patient?

- A Anti-glomerular basement membrane antibody
- B Anti-streptolysin O antibody
- C C3 nephritic factor
- D Hepatitis B surface antibody
- E HIV antibody

9 A 46-year-old woman has had worsening malaise for the past 36 hours. Her urine output is markedly diminished, and it has a cloudy brown appearance. On examination she has periorbital edema. Laboratory findings include serum creatinine of 2.8 mg/dL and urea nitrogen of 30 mg/dL. A renal biopsy is performed and on microscopic examination shows focal necrosis in glomeruli with glomerular basement membrane breaks and crescent formation. No immune deposits are identified with immunofluorescence. Which of the following autoantibodies is most likely detectable in her serum?

- A Anti-DNA topoisomerase antibody
- B Anti-glomerular basement membrane antibody
- C Anti-neutrophil cytoplasmic autoantibody
- D Antinuclear antibody
- E Anti-HBs Ag

10 A 44-year-old woman has developed a fever, nonproductive cough, and decreased urine output over the past 3 days. On physical examination, her temperature is 37.7° C, and blood pressure is 145/95 mm Hg. She has sinusitis. On auscultation, crackles are heard over all lung fields. A chest radiograph shows bilateral patchy infiltrates and nodules. The serum creatinine level is 4.1 mg/dL, and the urea nitrogen level is 43 mg/dL. The results of serologic testing are negative for ANA, but positive for C-ANCA. A renal biopsy specimen shows glomerular crescents and damage to small arteries. The result of immunofluorescence staining with anti-IgG and anti-C3 antibodies is negative. Which of the following additional microscopic findings is most likely to be seen in this biopsy?

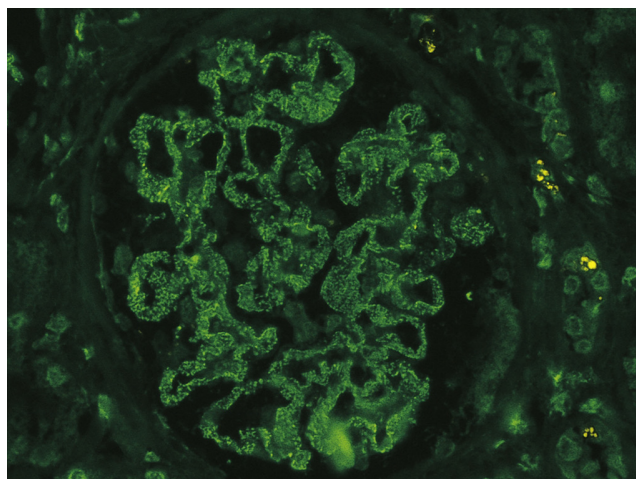
- A Focal segmental glomerulosclerosis
- B Glomerular basement membrane thickening
- C Hyperplastic arteriosclerosis
- D Infiltrations by neutrophils
- E Mesangial proliferation
- F Necrotizing granulomatous vasculitis

11 A 48-year-old man has had increased swelling in the extremities for 2 months. Physical examination showed generalized edema. A 24-hour urine collection yielded 4.1 g of protein (albumin and globulins). He did not respond to a course of corticosteroid therapy. A renal biopsy was done, and microscopic examination showed diffuse thickening of the basement membrane. Immunofluorescence staining with antibody to the C3 component of complement was positive in a granular pattern in the glomerular capillary loops. Two years later, he experiences increasing malaise. Laboratory studies now show serum creatinine level of 4.5 mg/dL and urea nitrogen level of 44 mg/dL. Which of the following immunologic mechanisms was most likely responsible for the glomerular changes observed in the biopsy specimen?

- A Antibodies that react with basement membrane collagen
- B Antibodies against streptococci that cross-react with the basement membrane
- C Cytotoxic T cells directed against renal antigens
- D Deposition of immune complexes on the basement membrane
- E Release of cytokines by inflammatory cells

12 A 7-year-old boy has become less active over the past 10 days. On physical examination, the boy has facial puffiness. Urinalysis shows no blood, glucose, or ketones, and microscopic examination shows no casts or crystals. The serum creatinine level is normal. A 24-hour urine collection yields 3.8 g of protein. He improves after corticosteroid therapy. He has two more episodes of proteinuria over the next 4 years, both of which respond to corticosteroid therapy. What is the most likely mechanism causing his disease?

- A Cytokine-mediated visceral epithelial cell injury
- B Cytotoxic T cell-mediated tubular epithelial cell injury
- C IgA-mediated mesangial cell injury
- D Immune complex-mediated glomerular injury
- E Verocytotoxin-induced endothelial cell injury

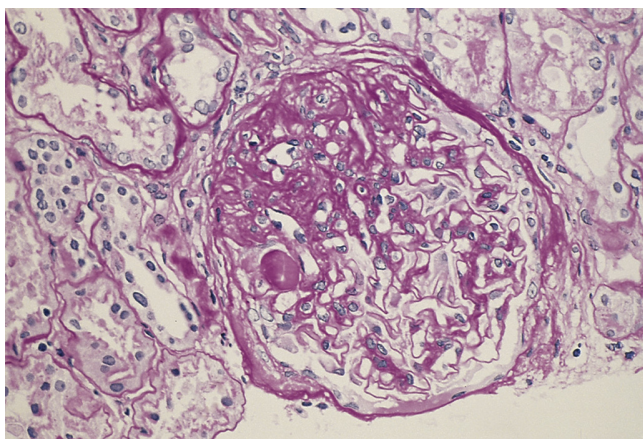


13 A 44-year-old man with increasing malaise for the past month now is bothered by increasing swelling in the hands and legs. On physical examination, there is generalized edema. He is afebrile, and his blood pressure is 140/90 mm Hg. Urinalysis shows a pH of 6.5; specific gravity 1.017; 4+ proteinuria; and no blood, glucose, or ketones. Microscopic examination of the urine shows no casts or RBCs and 2 WBCs per high-power field. The 24-hour urine protein level is 4.2 g. A renal biopsy specimen is obtained, and immunofluorescence staining with antibody to the C3 component of complement produces the pattern shown in the figure. Which of the following underlying disease processes is most likely to be present in this man?

- A Chronic hepatitis B virus infection
- B HIV infection
- C Multiple myeloma
- D Recurrent urinary tract infection
- E Nephrolithiasis

14 A 6-year-old girl has become increasingly lethargic over the past 2 weeks. On examination she has puffiness around the eyes. Her temperature is 36.9° C, and blood pressure is 100/60 mm Hg. Laboratory findings show serum creatinine, 0.7 mg/dL; urea nitrogen, 12 mg/dL; and cholesterol, 217 mg/dL. Urinalysis shows pH, 6.5; specific gravity, 1.011; 4+ proteinuria; lipiduria; and no blood or glucose. The 24-hour urine protein level is 3.8 g. The child's condition improves after glucocorticoid therapy. Which of the following findings by electron microscopy is most likely to characterize this disease process?

- A Areas of thickened and thinned basement membrane
- B Effacement of podocyte foot processes
- C Increased mesangial matrix
- D Reduplication of glomerular basement membrane
- E Subepithelial electron-dense humps



15 A 12-year-old girl has experienced increasing malaise for the past 2 weeks. On physical examination, she has periorbital edema. The child is afebrile. Laboratory findings show proteinuria on dipstick urinalysis, but no hematuria or glucosuria. Microscopic examination of the urine shows numerous oval fat bodies. The serum creatinine level is 2.3 mg/dL. She receives a course of corticosteroid therapy, but does not improve. A renal biopsy is performed and microscopic examination shows that approximately 50% of the glomeruli in the specimen are affected by the lesion shown in the figure. What is the most likely diagnosis?

- A Dense deposit disease
- B Focal segmental glomerulosclerosis
- C Minimal change disease
- D Nodular glomerulosclerosis
- E Postinfectious glomerulonephritis
- F Rapidly progressive glomerulonephritis

16 A 46-year-old Hispanic man has had increasing malaise with headaches and easy fatigability for the past 3 months. Physical examination reveals his blood pressure is 200/100 mm Hg. There are no palpable abdominal masses and no costovertebral tenderness. Laboratory studies show hemoglobin, 9.5 g/dL; hematocrit, 28.3%; MCV, 92 μm^3 ; creatinine, 4.5 mg/dL; and urea nitrogen, 42 mg/dL. Urinalysis reveals 3+ hematuria and 3+ proteinuria, but no glucose or leukocytes. A renal biopsy is done; light microscopic examination of the biopsy specimen shows that approximately 50% of the glomeruli appear normal, but the rest show that a portion of the capillary tuft is sclerotic. Immunofluorescence staining shows IgM and C3 deposition in these sclerotic areas. His history is significant for repeated episodes of passing dark brown urine, which failed to respond to corticosteroid therapy. Which of the following mechanisms is most likely responsible for his disease?

- A Attachment of anti-glomerular basement membrane antibodies
- B Deposition of immune complexes containing microbial antigens
- C Deposition of C3 nephritic factor (C3NeF)
- D Dysfunction of the podocyte slit diaphragm apparatus
- E Inherited defect in the basement membrane collagen

17 A 38-year-old woman has been feeling lethargic for 4 months. On physical examination, she is afebrile, and her blood pressure is 140/90 mm Hg. Laboratory findings show the serum creatinine level is 5.8 mg/dL. C3 nephritic factor is present in serum, resulting in hypocomplementemia, and the ANA test result is negative. Urinalysis shows 2+ blood and 1+ protein. A renal biopsy is done; microscopic examination shows hypercellular glomeruli and prominent ribbonlike deposits along the lamina densa of the glomerular basement membrane. Which of the following forms of glomerulonephritis is most likely to be present in this patient?

- A Chronic glomerulonephritis
- B Dense deposit disease
- C Membranous nephropathy
- D Postinfectious glomerulonephritis
- E Rapidly progressive glomerulonephritis

18 A 25-year-old man has a 5-year history of celiac sprue. Four days after a mild upper respiratory infection, he begins passing dark red-brown urine. The dark urine persists for the next 3 days and then becomes clear and yellow, only to become red-brown again 1 month later. There are no remarkable findings on physical examination. Urinalysis shows a pH of 6.5; specific gravity, 1.018; 3+ hematuria; 1+ proteinuria; and no glucose or ketones. Microscopic examination of the urine shows RBCs, but no WBCs, casts, or crystals. A 24-hour urine protein level is 200 mg. A renal biopsy specimen from the glomeruli of this patient is most likely to show which of the following alterations?

- A Diffuse cellular proliferation and basement membrane thickening
- B Granular staining of the basement membrane by anti-IgG antibodies
- C Mesangial IgA staining by immunofluorescence
- D Subepithelial electron-dense deposits
- E Thrombosis within the glomerular capillaries

19 One week after a mild flulike illness, a 9-year-old boy has an episode of hematuria that subsides within 2 days. One month later, he tells his parents that his urine is red again. On physical examination, there are no significant findings. Urinalysis shows a pH of 7; specific gravity, 1.015; 1+ proteinuria; 1+ hematuria; and no ketones, glucose, or urobilinogen. The serum urea nitrogen level is 36 mg/dL, and the creatinine level is 3.2 mg/dL. Serum electrophoresis shows increased IgA1. Which of the following glomerular structures is most likely to show structural alterations in this boy?

- A Basement membranes
- B Capillaries
- C Mesangium
- D Parietal epithelium
- E Podocytes

20 A 15-year-old boy has been passing dark-colored urine for the past month. On physical examination, he has bilateral sensorineural hearing loss and corneal erosions. Urinalysis shows a pH of 6.5; specific gravity, 1.015; 1+ hematuria; 1+ proteinuria; and no ketones, glucose, or leukocytes. The serum creatinine level is 2.5 mg/dL, and the urea nitrogen level is 24 mg/dL. A renal biopsy specimen shows tubular epithelial foam cells by light microscopy. By electron microscopy, the glomerular basement membrane shows areas of attenuation, with splitting and lamination of lamina densa in other thickened areas. What is the most likely diagnosis?

- A Mutation in a gene encoding type IV collagen
- B Increased synthesis of abnormal IgA
- C Autoimmune destruction of pancreatic beta cells
- D Acquired deficiency of ADAMTS13 metalloprotease
- E Toxic injury to slit diaphragm proteins

21 A 56-year-old woman is found on health screening to have a blood pressure of 168/109 mm Hg. No other physical examination findings are noted. Urinalysis shows a pH of 7.0; specific gravity, 1.020; 1+ proteinuria; and no blood, glucose, or ketones. The ANA and ANCA test results are negative. The serum urea nitrogen level is 51 mg/dL, and the creatinine level is 4.7 mg/dL. The hemoglobin A_{1c} concentration is within the reference range. An abdominal ultrasound scan shows bilaterally and symmetrically small kidneys with no masses. What is her most likely diagnosis?

- A Amyloidosis
- B Autosomal dominant polycystic kidney disease
- C Chronic glomerulonephritis
- D Microscopic polyangiitis
- E Nodular glomerulosclerosis

22 An autopsy study is performed involving persons with gross pathologic findings of bilaterally small kidneys (<100 g) that have a coarsely granular surface appearance. Microscopic examination shows sclerotic glomeruli, a fibrotic interstitium, tubular atrophy, arterial thickening, and scattered lymphocytic infiltrates. Which of the following clinical findings was most likely reported in these patients' medical histories?

- A Hemoptysis
- B Hypertension
- C Lens dislocation
- D Pharyngitis
- E Skin rash

23 A 33-year-old woman has had fever and increasing fatigue for the past 2 months. Over the past year, she has noticed soreness of her muscles and joints and has had a 4-kg weight loss. On physical examination, her temperature is 37.5° C, pulse is 80/min, respirations are 15/min, and blood pressure is 145/95 mm Hg. She has pain on deep inspiration, and a friction rub is heard on auscultation of the chest. Laboratory findings show glucose, 73 mg/dL; total protein, 5.2 g/dL; albumin, 2.9 g/dL; and creatinine, 2.4 mg/dL. Serum complement levels are decreased. CBC shows hemoglobin of 9.7 g/dL, platelet count of 85,000/mm³, and WBC count of 3560/mm³. A renal bi-

opsy specimen shows a diffuse proliferative glomerulonephritis with extensive granular immune deposits of IgG and C1q in capillary loops and mesangium. After being treated with immunosuppressive therapy consisting of prednisone and cyclophosphamide, her condition improves. Which of the following serologic studies is most likely to be positive in this patient?

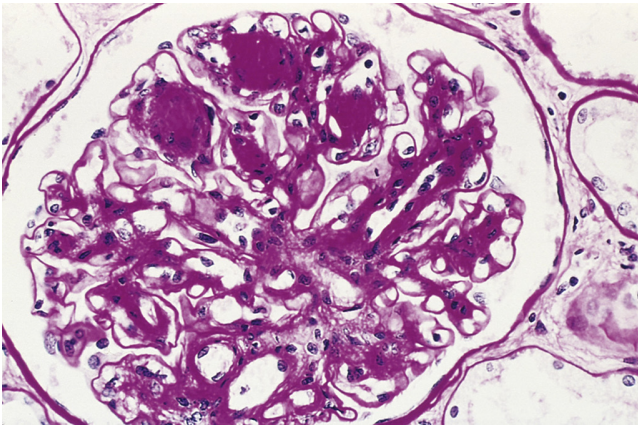
- A Anticentromere antibody
- B Anti-DNA topoisomerase I antibody
- C Anti-double-stranded DNA antibody
- D Anti-glomerular basement membrane antibody
- E Antihistone antibody
- F Antineutrophil cytoplasmic autoantibody
- G Antiribonucleoprotein

24 A 33-year-old woman with a history of intravenous drug use comes to the emergency department because she has had a high fever for the past 2 days. On physical examination, her temperature is 38.4° C. She has a palpable spleen tip, bilateral costovertebral angle tenderness, and diastolic cardiac murmur. Laboratory findings show a serum urea nitrogen level of 15 mg/dL. Urinalysis shows 2+ hematuria, and no glucose, protein, or ketones. A blood culture is positive for *Staphylococcus aureus*. Which of the following best describes the likely gross appearance of the kidneys in this patient?

- A Enlarged, and replaced by 1- to 4-cm, fluid-filled cysts
- B Marked pelvic and calyceal dilation with thinning of the cortices
- C Normal size, with smooth cortical surfaces
- D Shrunken, with uniformly finely granular cortical surfaces
- E Slightly swollen, with scattered petechial hemorrhages
- F Small and asymmetric, with irregular cortical scars and marked calyceal dilation
- G Wedge-shaped regions of yellow-white cortical necrosis

25 A 55-year-old woman with poorly controlled hyperglycemia for many years now has had burning pain on urination for the past 3 days. Physical examination shows a 2-cm ulceration on the skin of the heel and reduced sensation in the lower extremities. Her visual acuity is 20/100 bilaterally. Urinalysis shows 1+ proteinuria; 2+ glucosuria; and no blood, ketones, or urobilinogen. A urine culture contains more than 100,000 colony-forming units/mL of *Klebsiella pneumoniae*. Which of the following pathologic findings is most likely to be present in both her kidneys?

- A Deposits of IgG and C3 in the glomerular basement membrane
- B Effacement of podocyte foot processes
- C Formation of glomerular crescents
- D Mesangial deposits of IgA
- E Necrotizing granulomatous vasculitis
- F Nodular hyaline mesangial deposits



26 A 58-year-old man is found to have mild hypertension. Laboratory findings show a serum creatinine level of 2.2 mg/dL and urea nitrogen level of 25 mg/dL. Microalbuminuria is present, with excretion of 250 mg/day of albumin. Two years later, he remains hypertensive and has a serum creatinine level of 3.8 mg/dL, urea nitrogen level of 38 mg/dL, and 24-hour urine protein level of 2.8 g. A renal biopsy is done; the light microscopic appearance of a PAS-stained specimen is shown in the figure. Blood/serum test for which of the following is most likely to be abnormal in this patient?

- A Anti-glomerular basement membrane antibody
- B Antinuclear antibody
- C Anti-neutrophil cytoplasmic autoantibody
- D Anti-streptolysin O
- E C3 nephritic factor
- F Hemoglobin A_{1c}
- G Hepatitis B surface antigen

27 A 17-year-old boy is involved in a motor vehicle accident in which he sustains severe blunt trauma to the extremities and abdomen. Over the next 3 days, he develops oliguria and dark brown urine. The urine dipstick analysis is positive for myoglobin and for blood, but microscopic examination of the urine shows no RBCs. His serum urea nitrogen level increases to 38 mg/dL, and he undergoes hemodialysis for 3 weeks. His condition improves, but the urine output remains greater than 3 L/day for 1 week before the urea nitrogen returns to normal. Which of the following renal lesions was most likely present in this patient?

- A Acute pyelonephritis
- B Acute tubular injury
- C Malignant nephrosclerosis
- D Membranous nephropathy
- E Renal vein thrombosis

28 A 63-year-old man is in stable condition after an acute myocardial infarction when he became hypotensive for 3 hours before paramedical personnel arrived. Over the next week, the serum urea nitrogen level increases to 48 mg/dL, the serum creatinine level increases to 5 mg/dL, and the urine output decreases. He undergoes hemodialysis for the next 2 weeks and then develops marked polyuria, with urine output

of 2 to 3 L/day. His renal function gradually returns to normal. Release of which of the following substances most likely participated in the elevation of BUN, creatinine, and reduced urinary output?

- A Aldosterone
- B Endothelin
- C Erythropoietin
- D Natriuretic peptide
- E Vasopressin

29 A 19-year-old woman has had a fever and chills accompanied by right flank pain for the past 3 days. She has had two similar episodes during the past year. On physical examination, her temperature is 38.3° C, her blood pressure is 152/94 mm Hg, and there is right costovertebral angle tenderness. Laboratory findings show a serum glucose level of 77 mg/dL and creatinine level of 1 mg/dL. Urinalysis shows a pH of 6.5; specific gravity, 1.018; and no protein, blood, glucose, or ketones. Microscopic examination of the urine shows many WBCs and WBC casts. Which of the following is the most useful test to obtain on this patient?

- A Antinuclear antibody
- B Plasma renin
- C Renal biopsy
- D Urine culture
- E Abdominal CT scan

30 A 51-year-old woman has had dysuria and urinary frequency for the past week. On physical examination, her temperature is 38° C, and she has pain on palpation over the left costovertebral angle. Laboratory findings show glucose, 177 mg/dL; hemoglobin A_{1c}, 9.8%; hemoglobin, 13.1 g/dL; platelet count, 232,200/mm³; and WBC count, 11,320/mm³. Urinalysis shows a pH of 6.5; specific gravity, 1.016; 2+ glucosuria; and no blood, protein, or ketones. Microscopic examination of the urine shows numerous neutrophils, and a urine culture is positive for *Escherichia coli*. Which of the following complications is most likely to develop in this patient?

- A Acute tubular necrosis
- B Crescentic glomerulonephritis
- C Hydronephrosis
- D Necrotizing papillitis
- E Renal calculi

31 A 53-year-old woman has had fever and flank pain for the past 2 days. On physical examination, her temperature is 38.2° C, pulse is 81/min, respirations are 16/min, and blood pressure is 130/80 mm Hg. Urinalysis shows no protein, glucose, or ketones. The leukocyte esterase test is positive. Microscopic examination of the urine shows numerous polymorphonuclear leukocytes and occasional WBC casts. Which of the following organisms is most likely to be found in the urine culture?

- A *Cryptococcus neoformans*
- B *Escherichia coli*
- C Group A streptococcus
- D *Mycobacterium tuberculosis*
- E *Mycoplasma hominis*

32 A 30-year-old woman has had increasing malaise with nocturia and polyuria for the past year. She has had a high fever for the past 3 days. On physical examination, her blood pressure is 170/95 mm Hg. Urinalysis shows a pH of 7.5; specific gravity, 1.010; 1+ proteinuria; positive leukocyte esterase and nitrite; and no glucose, blood, or ketones. A renal ultrasound scan shows an enlarged right kidney with pelvic and calyceal enlargement and cortical thinning; the left kidney appears normal. A right-sided nephrectomy is performed, and grossly there are large U-shaped scars at the poles with underlying blunted calyces. Microscopic examination shows inflammatory infiltrates extending from the medulla to the cortex, with tubular destruction and extensive interstitial fibrosis. Lymphocytes, plasma cells, and neutrophils are abundant. Which of the following underlying conditions is most likely to produce these findings?

- A Polycystic kidney disease
- B Essential hypertension
- C Congestive heart failure
- D Systemic lupus erythematosus
- E Systemic amyloidosis
- F Vesicoureteral reflux

33 A 29-year-old woman has had a fever and sore throat for the past 3 days. On physical examination, her temperature is 38° C. The pharynx is erythematous, with yellowish tonsillar exudate. She is treated with ampicillin and recovers fully in 7 days. Two weeks later, she develops fever and a rash, and notices a slight decrease in urinary output. Her temperature is 37.7° C, and there is a diffuse erythematous rash on the trunk and extremities. Urinalysis shows a pH of 6; specific gravity, 1.022; 1+ proteinuria; 1+ hematuria; and no glucose or ketones. Microscopic examination of the urine shows RBCs and WBCs, including eosinophils, but no casts or crystals. What is the most likely cause of her disease?

- A Deposition of immune complexes with streptococcal antigens
- B Formation of antibodies against glomerular basement membrane
- C Hematogenous dissemination of septic emboli
- D Hypersensitivity reaction to ampicillin
- E Renal tubular cell necrosis caused by bacterial toxins

34 A 32-year-old man developed a fever and rash over 3 days. Five days later, he has increasing malaise. On physical examination, the maculopapular erythematous rash on his trunk has nearly faded away. His temperature is 37.1° C, and blood pressure is 135/85 mm Hg. Laboratory studies show a serum creatinine level of 2.8 mg/dL and blood urea nitrogen level of 29 mg/dL. Urinalysis shows 2+ proteinuria; 1+ hematuria; and no glucose, ketones, or nitrite. The leukocyte esterase result is positive. Microscopic examination of urine shows RBCs and WBCs, some of which are eosinophils. Which of the following most likely precipitated his renal disease?

- A Antibiotic ingestion
- B Congestive heart failure
- C Eating poorly cooked ground beef
- D Streptococcal pharyngitis
- E Urinary tract infection

35 A 72-year-old man with chronic arthritis has taken more than 3 g of analgesics per day, including phenacetin, aspirin,

and acetaminophen, for the past 20 years. He now has increasing malaise, nausea, and diminished mentation. On physical examination, his blood pressure is 156/92 mm Hg. Laboratory findings show serum urea nitrogen level of 68 mg/dL and creatinine level of 7.1 mg/dL. Which of the following renal diseases is he most likely to develop?

- A Acute tubular injury
- B Chronic glomerulonephritis
- C Hydronephrosis
- D Renal cell carcinoma
- E Renal papillary necrosis

36 A 49-year-old man is found on physical examination to have a blood pressure of 160/110 mm Hg, but no other abnormalities. Laboratory studies show serum glucose of 75 mg/dL, creatinine of 1.3 mg/dL, and urea nitrogen of 20 mg/dL. His plasma renin is elevated. CT angiography shows marked stenosis of his renal arteries. He is treated with an angiotensin-converting enzyme inhibitor. A week later, he has a headache for which he takes ibuprofen. Over the next day, his urine output decreases. This effect on urinary output is most likely mediated by the unopposed action of which of the following chemicals?

- A Aldosterone
- B Histamine
- C Nitric oxide
- D Prostaglandin
- E Tumor necrosis factor

37 A 28-year-old man is diagnosed with acute myelogenous leukemia (M2), with a total leukocyte count of 45,000/mm³, including 60% blasts. After induction with a multiagent chemotherapy protocol, he has an episode of lower abdominal pain accompanied by passage of red urine. He has no fever, dysuria, or urinary frequency. On physical examination, there are no remarkable findings. Urinalysis shows a pH of 5.5; specific gravity, 1.021; 2+ hematuria; and no protein, ketones, or glucose. There are no remarkable findings on an abdominal radiograph. Which of the following additional urinalysis findings is most likely to be reported for this patient?

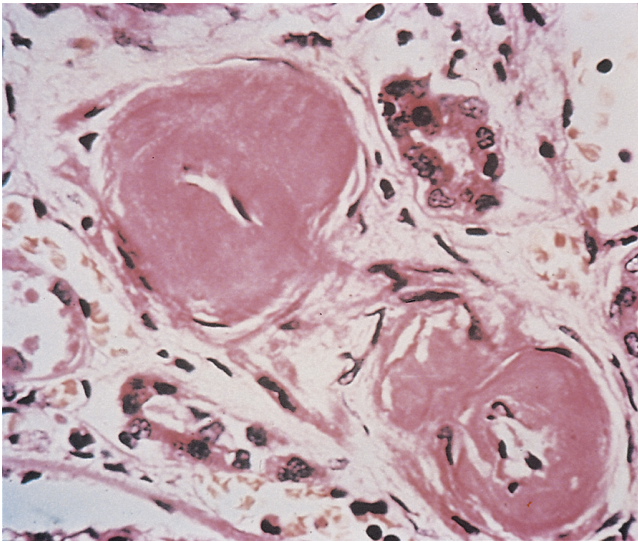
- A Bence Jones protein
- B Eosinophils
- C Myoglobin
- D Oval fat bodies
- E RBC casts
- F Uric acid crystals

38 A 49-year-old woman had a mastectomy of the right breast 2 years ago to remove a carcinoma. She now has bone pain, and a radionuclide scan shows multiple areas of increased uptake in the vertebrae, ribs, pelvis, and right femur. Urinalysis shows a specific gravity of 1.010, which remains unchanged after water deprivation for 12 hours. She undergoes multiple courses of chemotherapy over the next year. During this time, the serum urea nitrogen level progressively increases. Which of the following abnormal laboratory findings is most likely to be reported for this patient?

- A Hypercalcemia
- B Hypercholesterolemia
- C Hypergammaglobulinemia
- D Hyperglycemia
- E Hyperuricemia

39 A 63-year-old man has noted increasing back pain for 7 months. He has had three respiratory tract infections with *Streptococcus pneumoniae* during the past year. On examination, he has pitting edema to his thighs. Laboratory studies show total serum protein, 9.6 g/dL; albumin, 3.5 g/dL; creatinine, 3 mg/dL; urea nitrogen, 28 mg/dL; and glucose, 79 mg/dL. Urinalysis shows proteinuria of 4 g/24 hr, but no glucosuria or hematuria. Abdominal CT scan shows enlarged kidneys without cysts or masses. A renal biopsy specimen stained with H&E shows deposits of amorphous pink material within glomeruli, interstitium, and arteries. Which of the following diseases is he most likely to have?

- A Analgesic nephropathy
- B ANCA-associated granulomatous vasculitis
- C Type 2 diabetes mellitus
- D Membranous nephropathy
- E Multiple myeloma
- F Systemic lupus erythematosus



40 A 66-year-old woman has experienced five transient ischemic attacks within a week. On physical examination, the only abnormal finding is a blood pressure of 150/95 mm Hg. Urinalysis shows 1+ proteinuria, and no glucose, blood, or ketones. Microscopic examination of the urine shows no RBCs or WBCs and few oxalate crystals. On abdominal ultrasound, the kidneys are slightly decreased in size. The representative high magnification microscopic appearance of the kidneys is shown in the figure. Which of the following renal lesions is most likely to be present in this patient?

- A Acute tubular necrosis
- B Fibromuscular dysplasia
- C Hyaline arteriosclerosis
- D Interstitial nephritis
- E Necrotizing vasculitis

41 A 45-year-old man has had headaches, nausea, and vomiting that have worsened over the past 5 days. He has started "seeing spots" before his eyes and experienced periods of mental confusion. On physical examination, his blood pressure is 270/150 mm Hg. Urinalysis shows 1+ proteinuria; 2+ hematuria; and no glucose, ketones, or leukocytes. The serum urea nitrogen and creatinine levels are elevated. He dies 2 weeks later from a cerebral bleed. Which of the following histologic findings is most likely to be seen in this patient's kidneys at autopsy?

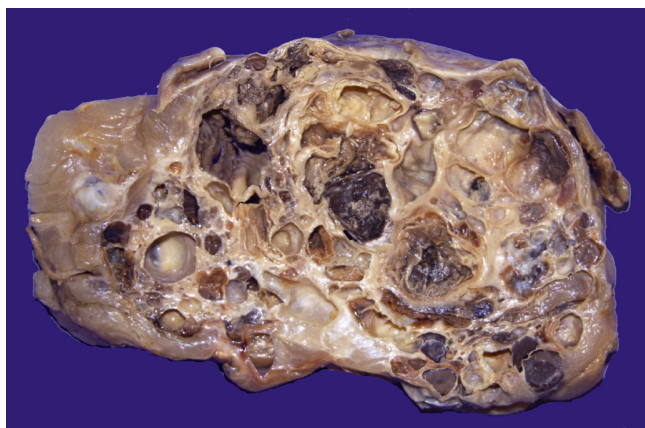
- A Glomerular crescents
- B Hyperplastic arteriosclerosis
- C Mesangial IgA deposition
- D Nodular glomerulosclerosis
- E Segmental tubular necrosis

42 A 5-year-old girl develops cramping abdominal pain and diarrhea 4 days after eating a hamburger, chili, and ice cream at a home barbecue. The next day, she has decreased urine output. On physical examination, there are petechial hemorrhages on the skin. Her temperature is 37° C, pulse is 90/min, respirations are 18/min, and blood pressure is 90/50 mm Hg. A stool sample is positive for occult blood. Laboratory findings show a serum creatinine level of 2.2 mg/dL and urea nitrogen level of 20 mg/dL. CBC shows hemoglobin, 10.8 g/dL; hematocrit, 32.4%; platelet count, 64,300/mm³; and WBC count, 6480/mm³. The peripheral blood smear shows schistocytes, and the serum D-dimer level is elevated. Urinalysis shows a pH of 6; specific gravity, 1.016; 2+ hematuria; and no protein or glucose. A renal biopsy specimen shows small thrombi within glomerular capillary loops. This complication develops most commonly after infection with which of the following organisms?

- A *Candida albicans*
- B *Clostridium difficile*
- C *Escherichia coli*
- D *Proteus mirabilis*
- E *Staphylococcus aureus*

43 Adult patients with bilateral renal cystic disease are found to have defects in renal tubular function. Their kidneys are markedly increased in size. They have ciliopathy, with reduced mechanosensing of tubular fluid flow. Mutations are found in a gene encoding for a protein component of these cilia. What is this protein most likely to be?

- A Elongin
- B Fibrocystin
- C Nephrocystin
- D Polycystin
- E Tuberin



44 A 61-year-old woman has experienced increasing malaise and bouts of abdominal pain for the past 5 years. On physical examination, she has a blood pressure of 150/95 mm Hg. On auscultation, a midsystolic click is heard over the mitral area. Bilateral abdominal masses are palpable. An abdominal ultrasound shows enlarged kidneys with the representative gross appearance shown in the figure. Which of the following extrarenal conditions is she most likely to have?

- A Berry aneurysm
- B Coagulopathy
- C Hepatic cirrhosis
- D Ischemic cardiomyopathy
- E Pulmonary fibrosis

45 A 20-year-old primigravid woman is in the third trimester and has felt minimal fetal movement. An ultrasound scan shows bilaterally enlarged echogenic kidneys and a markedly decreased amniotic fluid index. She gives birth to a stillborn male fetus at 33 weeks' gestation. At autopsy, there are deformations resulting from marked oligohydramnios, including flattening of the facies, varus deformities of the feet, and marked pulmonary hypoplasia. Microscopic examination of the liver shows multiple epithelium-lined cysts and a proliferation of bile ducts. Which of the following is the most likely renal disease in this fetus?

- A Autosomal dominant polycystic kidney disease
- B Autosomal recessive polycystic kidney disease
- C Medullary sponge kidney
- D Multicystic renal dysplasia
- E Urethral atresia

46 A 31-year-old woman experiences abdominal pain 1 week after noticing blood in her urine. She has had three episodes of urinary tract infection during the past year. There are no remarkable findings on physical examination. Urinalysis shows 2+ hematuria, 1+ proteinuria, hypercalciuria, and no glucose or ketones. Serum creatinine is 1.0 g/dL. Microscopic examination of the urine shows numerous RBCs and oxalate crystals. An abdominal CT scan with contrast shows linear striations radiating into the renal papillae, along with small cystic collections of contrast material in dilated collecting ducts. She is advised to increase her daily intake of fluids, and her condition improves. Which of the following renal cystic diseases is most likely to be associated with these findings?

- A Autosomal dominant polycystic kidney disease
- B Autosomal recessive polycystic kidney disease
- C Medullary sponge kidney
- D Multicystic renal dysplasia

47 A 6-year-old child has been drinking more water, with more frequent urination, for the past 7 months. On physical examination dehydration is noted. Urinalysis findings include pH of 6.5; specific gravity, 1.010; and no protein, blood, glucose, or ketones. There are no WBCs, RBCs, or casts. Serum electrolytes show Na^+ , 152 mmol/L; K^+ , 4.6 mmol/L; Cl^- , 120 mmol/L; HCO_3^- , 21 mmol/L; urea nitrogen, 29 mg/dL; and creatinine, 3.2 mg/dL. An ultrasound scan shows bilaterally small kidneys with barely visible medullary cysts concentrated at the corticomedullary junction. Which of the following genes is most likely mutated in this child?

- A *MCKD1*
- B *NPHP1*
- C *PKD1*
- D *PKHD1*



48 An 83-year-old man has experienced difficulty with urination for the past 15 years. On physical examination, he has a 2-cm nonhealing ulceration on the ball of his right foot. He has decreased sensation to light touch over his lower extremities. The representative gross appearance of his right kidney is shown in the figure. The left kidney has a similar appearance. Which of the following conditions is he most likely to have?

- A Analgesic abuse
- B Benign nephrosclerosis
- C Chronic pyelonephritis
- D Diabetes mellitus
- E Prostatic adenocarcinoma

49 A 42-year-old man has had right flank pain for the past 2 days. On physical examination, his temperature is 37.1° C, pulse is 70/min, respirations are 14/min, and blood pressure is 130/85 mm Hg. Laboratory studies show a serum creatinine level of 1.1 mg/dL. Urinalysis shows no blood, protein, or glucose, and microscopic examination of the urine shows no WBCs or RBCs. Abdominal CT scan shows a 7-cm eccentric lesion of the upper pole of the right kidney. The lesion is well circumscribed and cystic with a thin wall and focal hemorrhage. What is the most likely diagnosis?

- A Acute pyelonephritis
- B Acute tubular injury
- C Diabetic nephropathy
- D Hydronephrosis
- E Simple renal cyst
- F Renal cell carcinoma
- G Urothelial carcinoma

50 A 24-year-old man has severe lower abdominal pain that radiates to the groin. On a scale of 1 to 10, the pain is at 10 and comes in waves. He then notes blood in his urine. He has no underlying illnesses and has been healthy all his life. On physical examination, he is afebrile and has a blood pressure of 110/70 mm Hg. Laboratory studies show serum Na⁺, 142 mmol/L; K⁺, 4 mmol/L; Cl⁻, 96 mmol/L; CO₂, 25 mmol/L; glucose, 74 mg/dL; creatinine, 1.1 mg/dL; calcium, 9.1 mg/dL; and phosphorus, 2.9 mg/dL. Urinalysis shows a pH of 7; specific gravity of 1.020; and no protein, glucose, ketones, or nitrite. He is advised to drink more water. He likes iced tea and consumes large quantities over the course of a hot summer. He continues to have similar episodes. Which of the following substances is most likely to be increased in his urine?

- A Calcium oxalate
- B Cystine
- C Magnesium ammonium phosphate
- D Mucoprotein
- E Uric acid

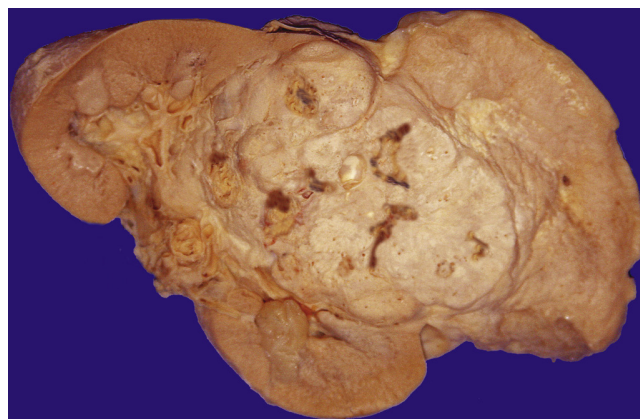
51 A 28-year-old, previously healthy man suddenly develops severe abdominal pain and begins passing red urine. There are no abnormalities on physical examination. Urinalysis shows a pH of 7; specific gravity, 1.015; 1+ hematuria; and no protein, glucose, or ketones. The patient is given a device to use in straining the urine for calculi. The next day, he recovers a 0.3-cm stone that is sent for analysis. The chemical composition is found to be calcium oxalate. What underlying condition is most likely to be present in this man?

- A Acute bacterial cystitis
- B Diabetes mellitus
- C Idiopathic hypercalciuria
- D Primary hyperparathyroidism
- E Secondary gout

52 A 54-year-old woman has had recurrent urinary tract infections for the past 15 years. On many of these occasions, *Proteus mirabilis* was cultured from her urine. For the past 4 days, she has had a burning pain on urination and urinary frequency. On physical examination, her temperature is 37.9° C, pulse is 70/min, respirations are 15/min, and blood pressure is 135/85 mm Hg. There is marked tenderness on deep pressure over the right costovertebral angle and on deep ab-

dominal palpation. Urinalysis shows a pH of 7.5; specific gravity, 1.020; 1+ hematuria; and no protein, glucose, or ketones. Microscopic examination of the urine shows many RBCs, WBCs, and triple phosphate crystals. Which of the following renal lesions is most likely to be present?

- A Acute tubular injury
- B Malignant nephrosclerosis
- C Papillary necrosis
- D Renal cell carcinoma
- E Staghorn calculus



53 A 55-year-old man has had back pain and has passed dark-colored urine for the past month. On physical examination, there is tenderness over the right costovertebral angle. Urinalysis shows a pH of 6; specific gravity, 1.015; 2+ hematuria; and no protein, glucose, or ketones. Microscopic examination of the urine shows numerous RBCs, few WBCs, and no casts or crystals. The figure shows the representative gross appearance of the renal lesion. Which of the following substances is most likely to be increased in the blood of this man?

- A Cortisol
- B Globulins
- C Erythropoietin
- D Renin
- E Vasopressin

54 A 60-year-old man has a feeling of fullness in his abdomen and a 5-kg weight loss over the past 6 months. He has a 50 pack-year smoking history. Physical examination is normal. Laboratory studies show hemoglobin of 8.3 g/dL, hematocrit of 24%, and MCV of 70 μm^3 . Urinalysis shows 3+ hematuria, but no protein, glucose, or leukocytes. Abdominal CT scan shows an 11-cm mass in the upper pole of the right kidney. A right-sided nephrectomy is performed, and gross examination reveals that the mass has invaded the renal vein. Microscopic examination of the mass shows cells with abundant clear cytoplasm. Which of the following molecular abnormalities is most likely to be found in tumor cell DNA?

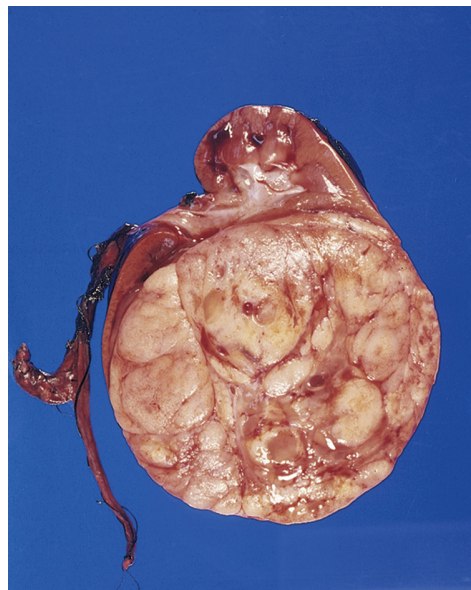
- A Homozygous loss of the von Hippel-Lind (VHL) gene
- B Integration of human papillomavirus type 16 (HPV-16)
- C Microsatellite instability
- D Mutational activation of the *MET* proto-oncogene
- E Trisomy of genes on chromosome 7

55 Members of a family with a history of renal cancers undergo ultrasound screening. Two adults are found to have multifocal and bilateral renal mass lesions. Biopsies are obtained, and microscopic examination shows a papillary pattern. A mutation involving which of the following genes is most likely to be found in this family?

- A *MET*
- B *PKD1*
- C *RAS*
- D *TSC1*
- E *WT1*

56 A 60-year-old man has noted a nonproductive cough along with back pain for 4 months. He has passed darker urine for 1 month. He has a 50 pack/year history of smoking. On examination, his blood pressure is 175/110 mm Hg. He has tenderness to percussion of the upper back. Urinalysis shows 3+ blood but no casts or crystals. Chest CT imaging shows a 4-cm solid nodule in the right lower lobe of his lung, as well as 1- to 2-cm lytic lesions in thoracic vertebrae. A neoplasm is most likely to have arisen in which of the following urinary tract locations in this man?

- A Bladder dome
- B Calyx
- C Penile urethra
- D Renal cortex
- E Urachus
- F Ureter



57 A 4-year-old girl has complained of abdominal pain for the past month. On physical examination, she is febrile, and palpation of the abdomen shows a tender mass on the right side. Bowel sounds are present. Laboratory studies show hematuria without proteinuria. Abdominal CT scan shows a 12-cm, circumscribed, solid mass in the right kidney. A right-sided nephrectomy is performed; the gross appearance of the mass is shown in the figure. What is the most likely diagnosis?

- A Angiomyolipoma
- B Interstitial cell tumor
- C Renal cell carcinoma
- D Transitional cell carcinoma
- E Wilms tumor

ANSWERS

1 A He has azotemia, and the BUN-to-Cr ratio is >20:1, the low FE_{Na} , high urine specific gravity, and hypotension suggest prerenal azotemia, and cardiac failure is the most likely cause. Membranous nephropathy is a renal cause with BUN-to-Cr ratio of 10:1 or less and likely proteinuria at nephrotic levels. Prostatic hyperplasia with urinary tract obstruction may produce postrenal azotemia with BUN-to-Cr ratio between 10:1 and 20:1. Systemic lupus erythematosus is likely to produce nephritic syndrome with RBCs and RBC casts, and BUN-to-Cr ratio of 10:1 or less. Urothelial carcinoma is likely to be unilateral, with sufficient reserve renal function in the remaining kidney to prevent azotemia.

PBD9 898–899 BP9 517–518 PBD8 906–907 BP8 542

2 C The findings fit with immune-mediated glomerular injury with antigen-antibody complex deposition, typical of an underlying disease such as SLE, and nephritic picture. Type I hypersensitivity may play a role in drug-induced acute interstitial nephritis. Type II hypersensitivity is present

with anti-glomerular basement membrane diseases, such as Goodpasture syndrome. Type IV hypersensitivity plays a role in renal transplant rejection and some cases of drug-induced interstitial nephritis.

PBD9 904–906 BP9 519–522 BPD8 912–915 BP8 546–548

3 B The appearance of membranoproliferative glomerulonephritis, a cause for nephritic syndrome in adults, is described. Mesangial cells have a phagocytic function, but they also can elaborate inflammatory mediators, cytokines, and matrix. Proliferation of mesangial cells may be induced by injury, particularly immune complex deposition. Juxtaglomerular cells secrete renin. Parietal epithelial cells line the Bowman capsule and may proliferate with severe glomerular injury to produce crescents. Podocytes are visceral epithelial cells that form the filtration barrier. Endothelial cells are most likely to be damaged with thrombotic microangiopathies.

PBD9 900–902 BP9 527–528 PBD8 909–911 BP8 548–549